OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 1 Examinations 2025-2026

Course Instance Code(s)	4BCT1, 1MEEE1,1MECS1, 1CSD1, 1ACS1, 2SPE1, 1MAI1, 1MEES1, 1MECS1
Exam(s)	B.Sc. Degree (Computer Science and Information Technology) B.E. Energy Systems Engineering, MSc in Mechanical Engineering, ME (Electrical and Computer Engineering), M.Sc. in Computer Science (Data Analytics), M.Sc. in Computer Science (Adaptive Cybersecurity), M.Sc. in Computer Science (Artificial Intelligence), Structured Ph.D. (Engineering), ME (Computer Science and Information Technology)
Module Code(s)	CT561
Module(s) **Bonded**	Systems Modelling and Simulation No [☑] Yes [ ] List bonds
Paper No. **External Examiner(s)** **Internal Examiner(s)**	1 Dr. Josh Murphy Dr. Enda Howley *Prof. Jim Duggan

Instructions: Answer 3 questions out of 4. All questions will be marked equally.

Duration 2 hours

No. of Pages 5

Discipline(s) Computer Science

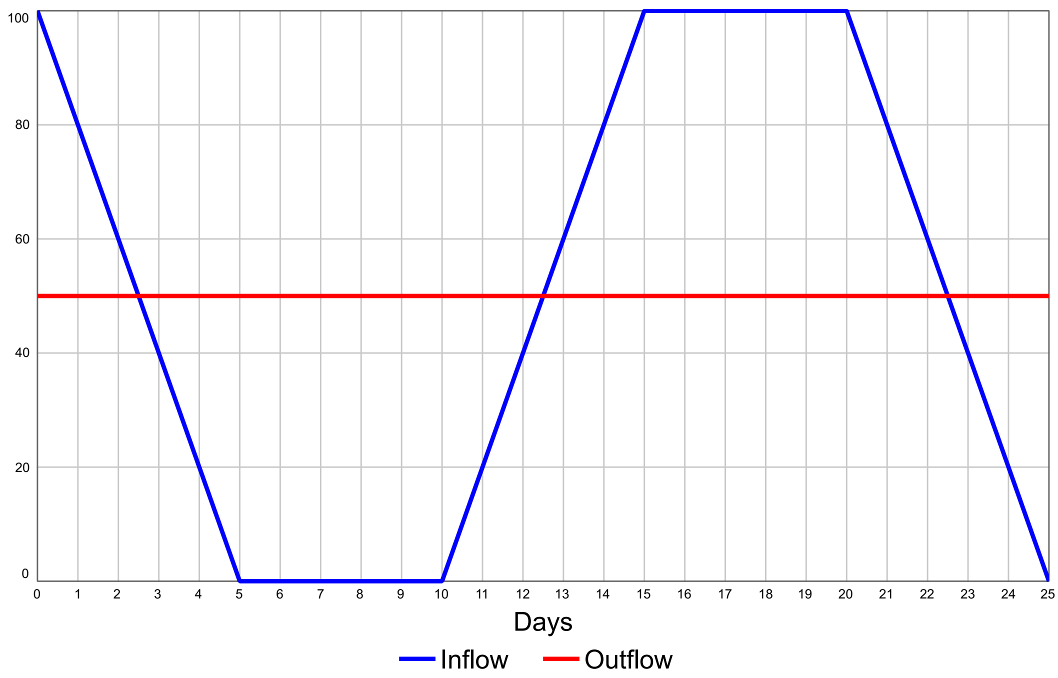
Requirements:

Remove Paper from exam venue, publish online	No [ ]	Yes[☑]
MCQ Answer sheet	No [☑]	Yes [ ]
Handout	No [☑]	Yes [ ]
Formulae & Log Tables**	No [☑]	Yes [ ]
Cambridge Tables 2nd Edition**	No [☑]	Yes [ ]
Graph Paper*** A4 Graph Paper 1mm 0.1cm Squared (Standard)	No [ ]	Yes [☑]
Other Materials	No [☑]	Yes [ ]
Graphic material in colour	No []	Yes [☑]

End of requirements

1. (a) Consider the two flow variables shown in the chart below. The inflow is coloured blue, while the outflow is a constant 50 units per day for the entire simulation, and coloured red.

    1. Draw a stock and flow model (include the net flow).
    2. Calculate and show the net flow.
    3. Use graphical integration to plot the stock over time, clearly explaining the different behaviour modes for each time interval. Assume the stock starts at 100 units. Note that it is ok if the stock goes negative.



[12]

(b) Describe Euler's method of integration, and state the underlying assumptions.

Under what conditions will Euler's method overestimate the true value of a stock?

Support your answer by considering the numerical integration of the net flow over the time interval 0-2.5 from the model in part (a), assuming DT=0.5.

[5]

(c) Based on the flows shown in part (a), calculate the stock using Euler's equation where DT=5.

[8]

2. (a) Show, for a stock S, how its fractional decrease flow (Outflow) can be modified to allow for a varying decrease fraction (G). Assume the decrease fraction is itself an adjustment to a goal structure, where G^* is the maximum decrease fraction, and AT is the adjustment constant.

Comment on how this structure could be used in a real-world modelling context.

[8]

(b) Based on the SIR model, add two policy structures that can be used to experiment with countermeasures. These model assumptions are:

- A quarantine stock where a proportion (20%) of susceptible people (governed by the force of infection) are moved to quarantine, while the remaining 80% will move to the infected stock. Those in quarantine recover at the same rate (i.e. 2 days) as those in the infected stock.

- A vaccinated stock, where (only) susceptible people move through a fractional decrease flow, where the flow is governed by the fraction *vaccination fraction*. This *vaccination fraction* starts at 0, and moves to its maximum (0.15) via a goal seeking trajectory.

- The value for $R_0 = 2$ (where R_0 is the average number of infections generated by an infectious person in a totally susceptible population).

- The total population N = 10,000, with one of these initially infected. Assume that people in quarantine do not infect others.

- The model should contain the overall attack rate, which is the proportion of people who have been infected.

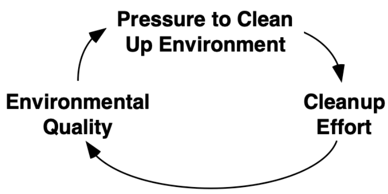
All equations must be specified, and feedbacks shown.

[12]

(c) Sketch – and explain - what you think would be the relationship between the *quarantine fraction* (x-axis, assume 0, 0.05, 0.10, 0.15)), and the *final vaccination fraction* (y-axis assume 0.0, 0.05, 0.10, 0.15), using different sized circles to represent the magnitude of the final attack rate at different combinations of the two parameters.

[5]

3. (a) Use the following example to define the terms link polarity, loop polarity, and feedback. Discuss what the behaviour this model will generate.



[6]

(b) Construct a stock and flow model from the following description of an insurance claims process workload. Clearly show the link polarities and loop polarities.

There is a single Stock (**Claims Backlog**) with two inflows (**New Claims** and **Reworked Claims**) and one outflow (**Claims Completed**).

1. **Claims Backlog** is increased by the inflows, and reduced by the outflow.
2. As **Claims Backlog** increases, so too does **Work Pressure**.
3. There are three policies use in response to an increase in **Work Pressure**
    - **Time per Claim** is decreased, which leads to an increase in **Claims Completed.**
    - **Temporary Staff** are increased, which leads to an increase in **Claims Completed**.
    - The **Workweek** is increased which leads to an increase in **Claims Completed**.
4. These three policies also have side-effects.
    - A decrease in **Time Per Claim** leads to an increase in **Reworked Claims**.
    - An increase in **Temporary Staff** leads to an increase in **Time Per Claim**.
    - An increase in **Workweek** leads to an increase in **Fatigue**, and this causes a decrease in **Claims Completed.**

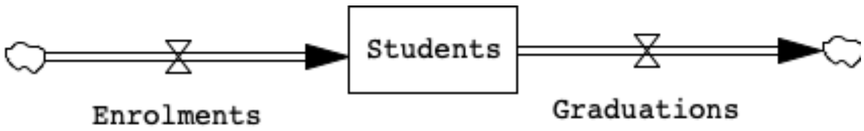
[12]

(c) Based on the stock and flow model from part (b):

- Identify the three policy loops and the three side-effects loops.
- Discuss what your think is the best course of action for coping with an increase in workload.

[7]

4. (a) For the following stock and flow model, add a stock and flow structure (with equations) that will model expected enrolments of students to the University.



[5]

(b) Explain how the *adjustment to a goal* structure can be extended to form the stock management structure. What is the key difference between both structures?

When explaining the stock management structure describe: (1) the anchor and (2) the adjustment.

[5]

(c) Create a stock and flow model (with all equations) of staff recruitment for a pharmaceutical manufacturing company, where the hire rate is based on the stock management structure.

Show all feedbacks in the model.

Assume there are three types of employee (Trainee, Experienced and Senior), and that attrition can happen at each stage.

The target number of employees is 300

There are 100 employees in each category

The progression delay is 12 months (Trainee-Experienced) and 36 months (Experienced-Senior).

The attrition rate is 5% for Trainees, 3% for Experienced, and 1% for Senior.

[10]

(d) The model defined in part (c) is in steady state (with 300 employees in total). Sketch what would happen if the target number of employees dropped to 100. How would the equations react in order to ensure that the new target is reached?

[5]

END